ACI NOTES - PART 5

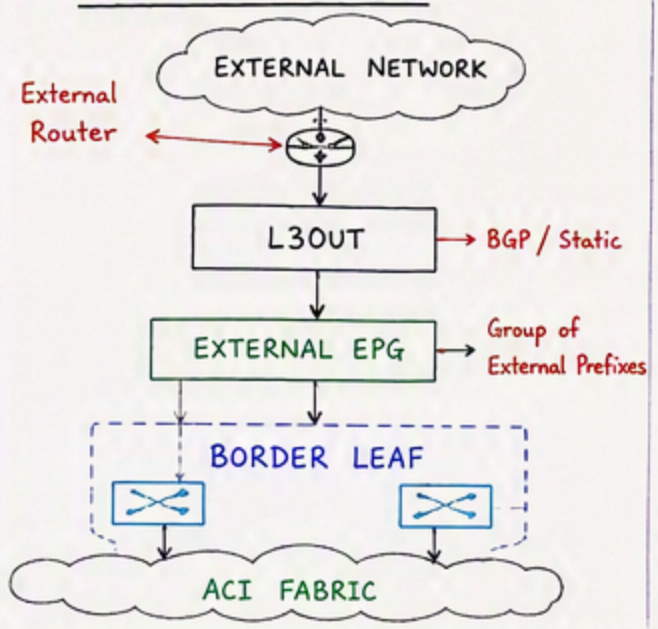
L3OUT DEEP DIVE & ROUTE CONTROL

CONNECTING ACI FABRIC TO EXTERNAL NETWORKS

1. WHAT IS L3OUT ?

- L3Out = Connection between ACI Fabric and External Network.
- Provides Routing between internal VRFs and external networks.
- Uses External EPG for grouping external prefixes.
- Connected on Border Leaf.

2. L3OUT COMPONENTS



3. L3OUT PACKET FLOW (INTERNET EXAMPLE)

1. Host 10.10.10.10 wants to reach 8.8.8.8
2. Host ARPs for Gateway 10.10.10.1 (Anycast Gateway on Leaf)
3. Leaf does VRF Route Lookup
4. Matches 0.0.0.0/0 → External EPG-INTERNET
5. Contract Check (Source EPG → Ext EPG)
6. Service Graph? (PBR Redirect)
7. Traffic sent to Border Leaf
8. Border Leaf forwards to External Router
9. Internet Response comes back same path

4. EXTERNAL EPG

- Represents external prefixes.
- Examples:
  - EPG-INTERNET → 0.0.0.0/0
  - EPG-MPLS → 172.16.0.0/12
  - EPG-PARTNER → 10.200.0.0/16
- Used in Contracts.
- Attached to L3Out.

5. ROUTE CONTROL PROFILE (RCP)

- Controls which routes are Imported / Exported.
- Applied to L3Out.
- Prevents unwanted routes from entering/exiting fabric.
- Types : Import / Export / Shared

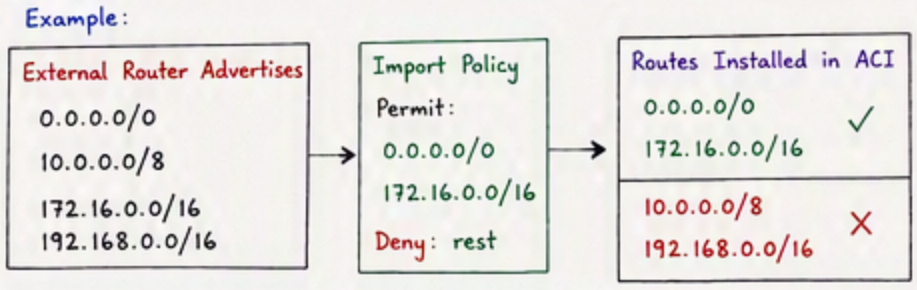
6. ROUTE CONTROL DIRECTION



7. IMPORT ROUTE CONTROL

Controls routes learned from external router.
Only permitted routes are installed in ACI.

Example:



8. EXPORT ROUTE CONTROL

Controls which ACI routes are advertised outside.
Prevents advertising internal/ sensitive networks.

Example:

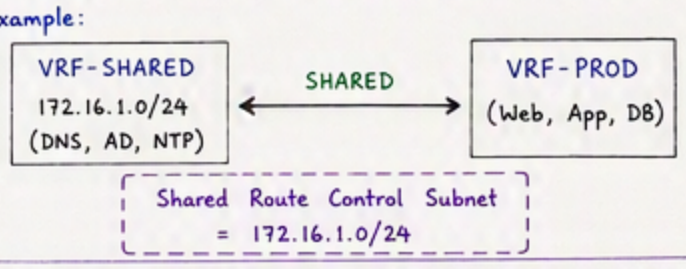


9. SHARED ROUTE CONTROL

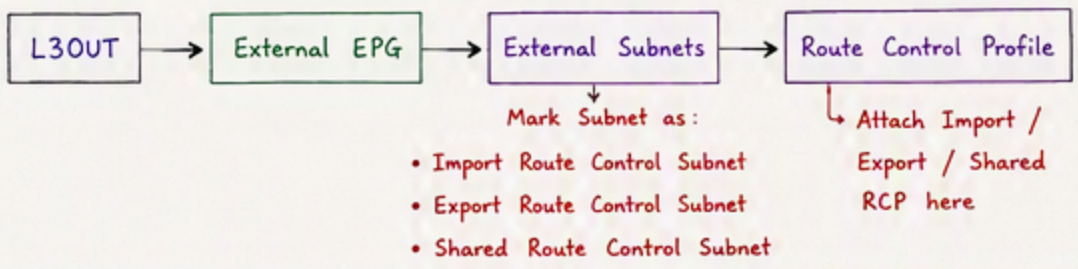
Used for Route Leaking / Shared Services.

Allows specific routes to be shared between VRFs or L3Outs.

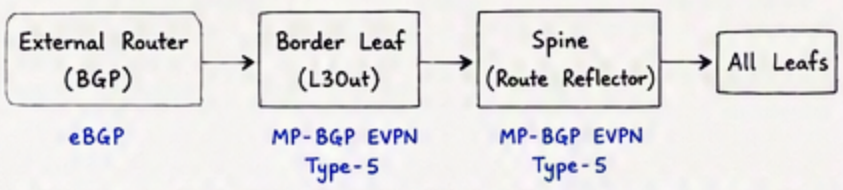
Example:



10. ROUTE CONTROL ATTACHMENT POINTS



11. ROUTE PROPAGATION (HOW ROUTE REACHES ALL LEAFS)



Result: All Leafs learn external route via MP-BGP EVPN (Type-5).

12. COMMON TROUBLESHOOTING STEPS

1. Check L3Out Interfaces & Adjacency (show ip interface brief vrf <vrf>)
2. Check Route on Border Leaf (show ip route vrf <vrf>)
3. Check Route Control Profile (Import/Export/Shared)
4. Check if route is redistributed into EVPN (show bgp l2vpn evpn route-type 5)
5. Check if route exists on Spine (show bgp l2vpn evpn summary)
6. Check if route exists on remote Leaf
7. Check Contracts between Source EPG and External EPG
8. Check Service Graph / PBR if configured
9. Check External Subnets flags (Import/Export/Shared)

13. COMMON ISSUES

1. Route present on Border Leaf but not on Leafs → RCP / Redistribution issue
2. Traffic drops after route leak → Missing Contract
3. Can't reach Internet → No default route / Wrong RCP / L3Out down
4. Route exported but not visible externally → Export RCP / BGP issue
5. Asymmetric routing → Gateway migration not done properly
6. Missing External EPG → Route has no destination EPG match

14. MEMORY AID

Import = What comes INTO ACI
Export = What goes OUT of ACI
Shared = What is shared BETWEEN
Type-2 = Endpoint (MAC + IP)
Type-5 = Prefix / Network
COOP = Where is the endpoint?
MP-BGP = How do I reach the network?

15. QUICK CHECK CLI

- show ip route vrf <vrf>
- show ip bgp summary vrf <vrf>
- show bgp l2vpn evpn route-type 5
- show ip route vrf <vrf> detail
- show zoning-rule
- show contracts usage
- show l3out
- show l3out route-control
- show ip interface brief vrf <vrf>

KEY TAKEAWAY: Route Control Profiles give you control, security and scalability in. They decide what enters, what leaves and what is access the fabric.

ROUTING + POLICY = COMMUNICATION ☺